

ELEC9731, 2023, Term 1, Take Home Exam

due – Monday 01/05/2023 17:00 (Sydney time).

submission – please submit as **pdf** file by email
(a.savkin@unsw.edu.au)
use subject: ELEC 9731- Take Home Exam

Late assignments will be penalised at 10% of the maximum value per day late

Question 1 (16 marks, Dorfman, Bishop). Consider the control system of Fig. 12.1 of Lecture Notes with

$$G(s) = \frac{3}{s(s^2 + 4s + 5)}.$$

Design a PID controller to achieve

- (a) an acceleration error constant $K_a = 2$;
- (b) a phase margin equal to $\frac{\pi}{4}$;
- (c) a bandwidth greater than 2.8 rad/s.

Select an appropriate pre-filter and plot the response to a step input.

Reminder: acceleration error constant is the constant K_a such that the steady state error to a parabolic input, $r(t) = 0.5t^2$ is equal to K_a^{-1} .

Question 2 (18 marks). Consider the following scalar discrete-time control system

$$x(k+1) = 4x(k) + u(k)$$

where $x \in \mathbf{R}$, $u \in \mathbf{R}$ and $x(0) = 2$. Let $N > 1$ be some integer.

- (a) Consider the following cost function

$$J = \sum_{k=0}^N u(k)^2.$$

Find the optimal control law and the minimal cost.

- (b) Consider the following cost function

$$J = \sum_{k=0}^{+\infty} u(k)^2.$$

Find the optimal control law and the minimal cost.

(c) Consider the following cost function

$$J = \sum_{k=0}^N x(k)^2.$$

Find the optimal control law and the minimal cost.

Question 3 (16 marks). Consider the continuous-time linear system with the disturbance input $w(t)$ and the measured output $y(t)$ where the transfer function $T(s)$ from w to y is

$$T(s) = \frac{s(s+1)}{(s+2)(s+3)(s^2+6s+10)}.$$

Also, we assume that there is a measurement noise input $v(t)$.

- (a) Obtain a state-space model for this system.
- (b) Is this system stable?
- (c) Assume that $\dot{w}(t)$ and $\dot{v}(t)$ are white-noise processes. Take some sensible impulsive correlation matrices Q and R and generate a Kalman filter for estimating the states of the system.
- (d) Generate an H_∞ filter to estimate the states of the system.
- (e) Simulate the plant and the filters from (c) and (d) for various types of $w(t)$ and $v(t)$, e.g. white noises, sinusoidal inputs, some other deterministic functions. Compare the performance of the filters.